Docker #1

UNDER THE HOOD

By Kiung Jung

Google Developer Student Clubs
Yonsei Univ Seoul Campus

1. 시스템 구조 (네트워크 연결)

2. 시스템 구조 (VM, Docker)

3. docker download

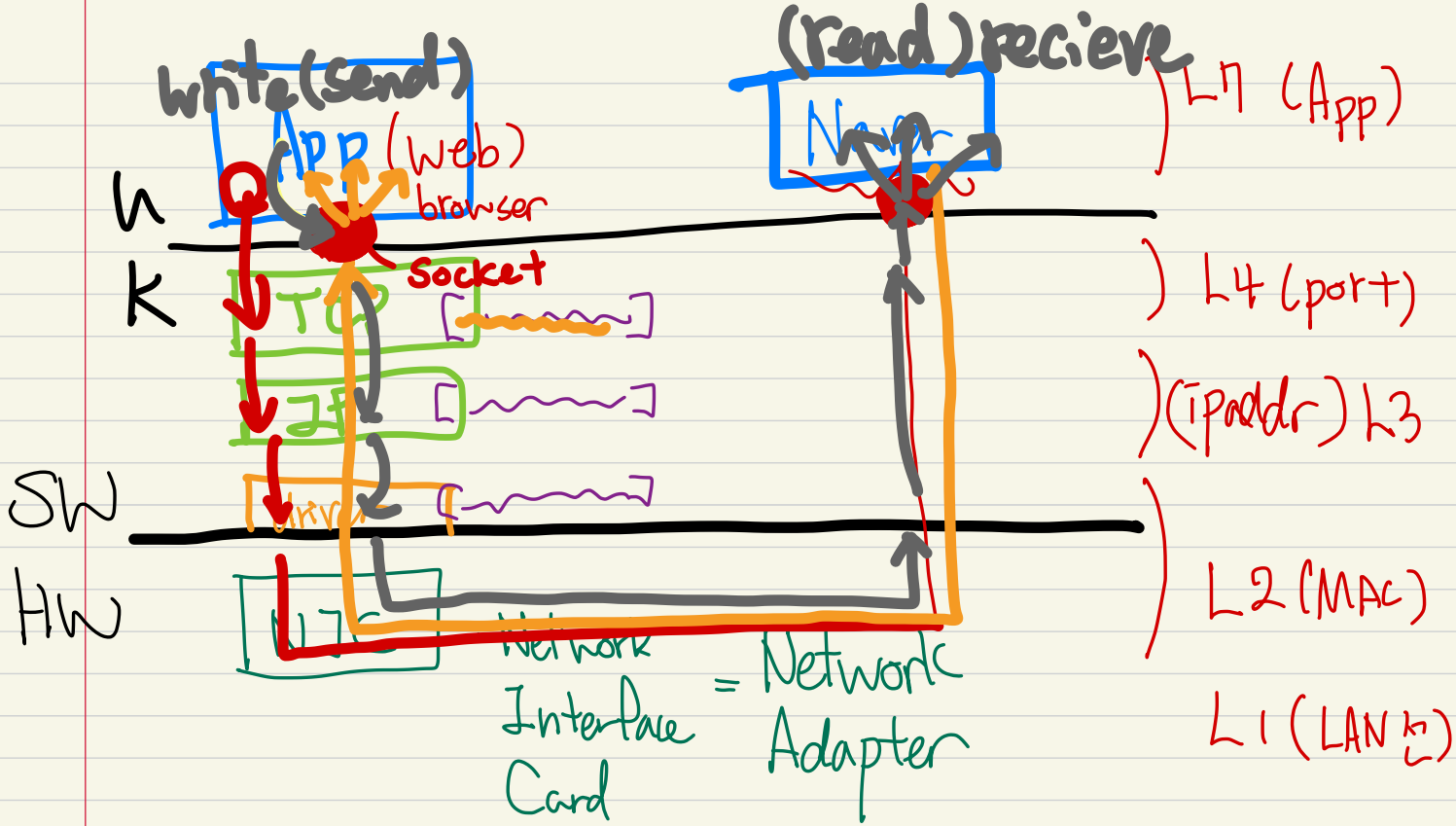
"docker group add"

3-1. docker registry

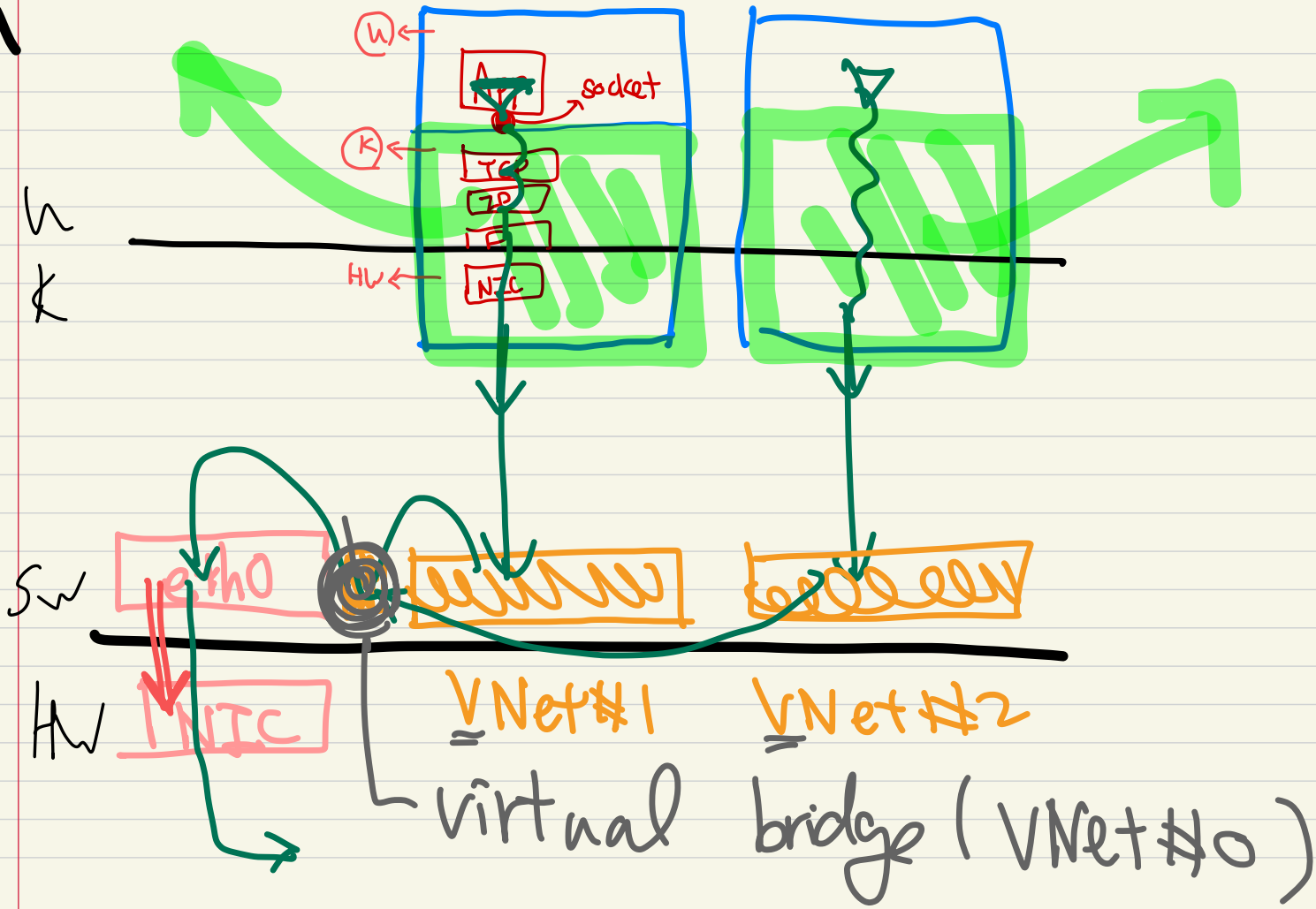
pull / push

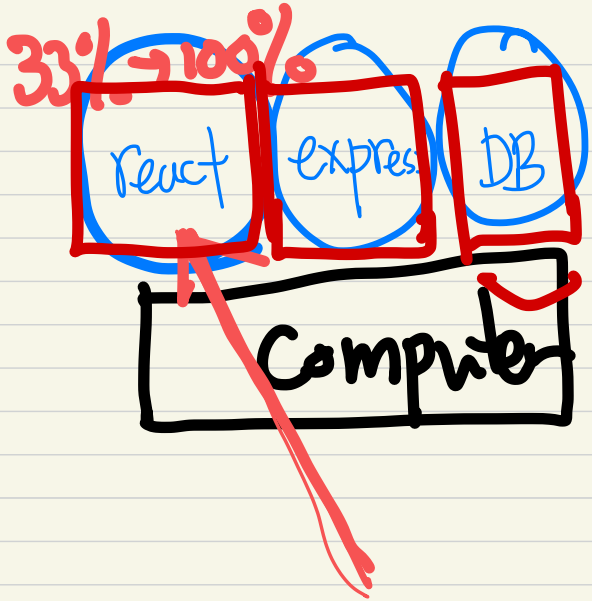
4. docker image layer

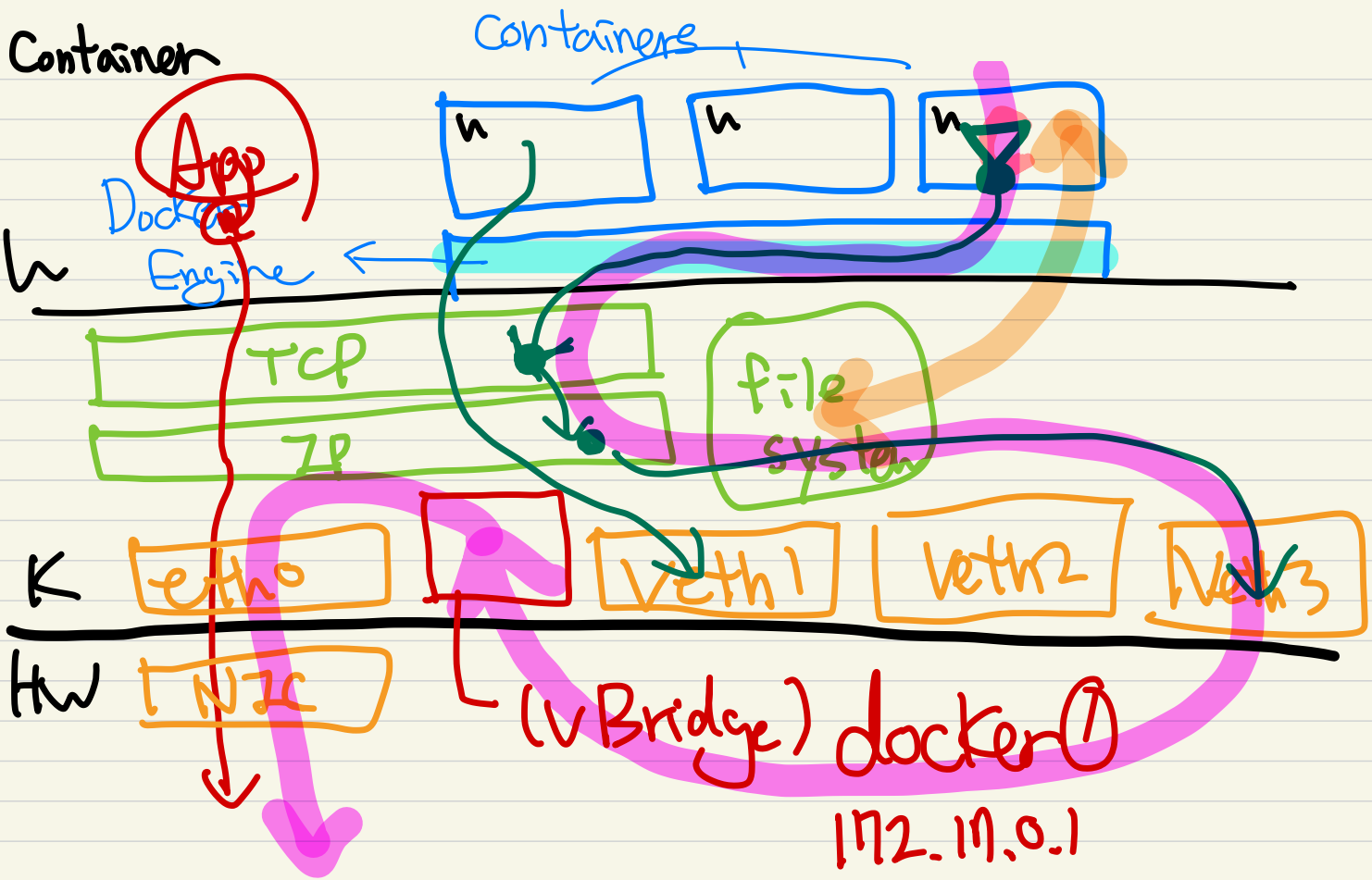
5. docker context


write(send)
App (web)
browser
(read) recieve
Naver
L7 (App)
Socket
TCP
IP
driver
L4 (port)
(ipaddr) L3
SW
HW
NIC
L2 (MAC)
Network Interface Card
= Network Adapter
L1 (LAN 선)

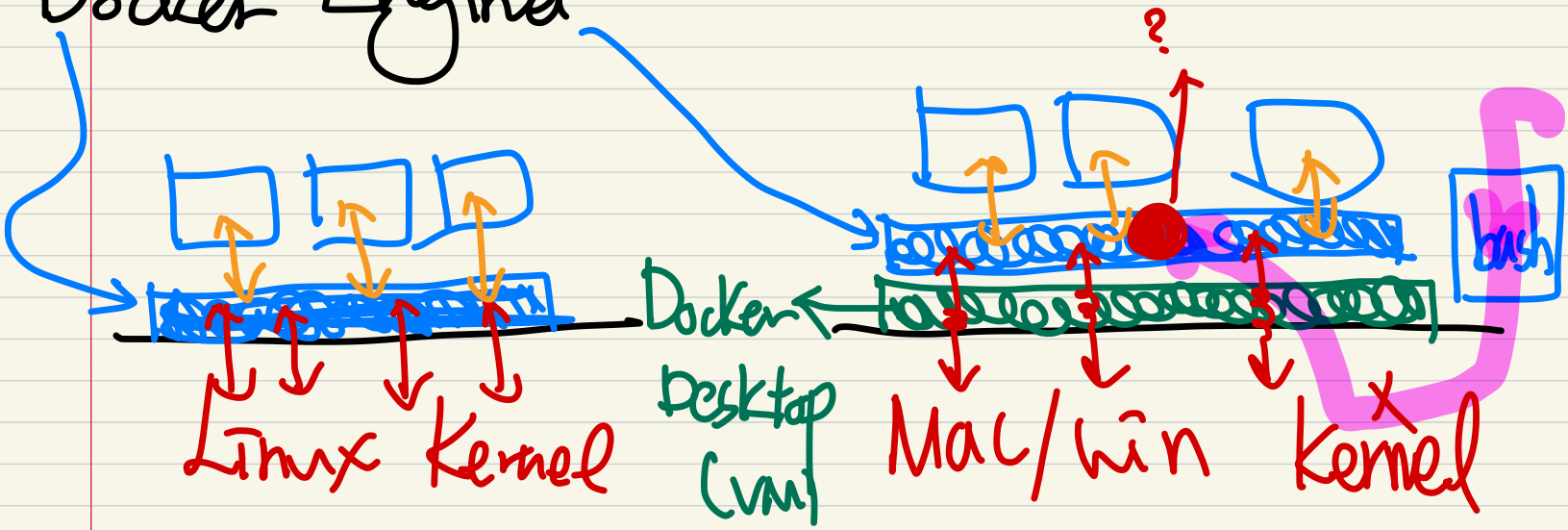
VM


W
socket
K
TCP
IP
HW
NIC
h
k
sw
eth0
HW
NIC
VNet#1
VNet#2
virtual bridge (VNet#0)


33% → 100%
react
express
DB
Computer


Container
App
Containers
Docker
Engine
U
TCP
IP
file
system
K
eth0
veth1
veth2
veth3
HW
NIC
(vBridge) docker0
172.17.0.1

Docker Engine



Pull & Push

for what? image ⟷ container

붕어빵틀 → 실행 → 붕어빵

Docker engine

- docker
  - push
  - pull ubuntu (dockerhub)
    - Github Packages AWS ECR
    - Local Registry GCP CR

Dockerhub

- pub img : 무제한
- pri img : 1 only

→ AWS ECR

Img

- docker pull ubuntu(:latest)

1GB

:22.04

1GB

:20.04

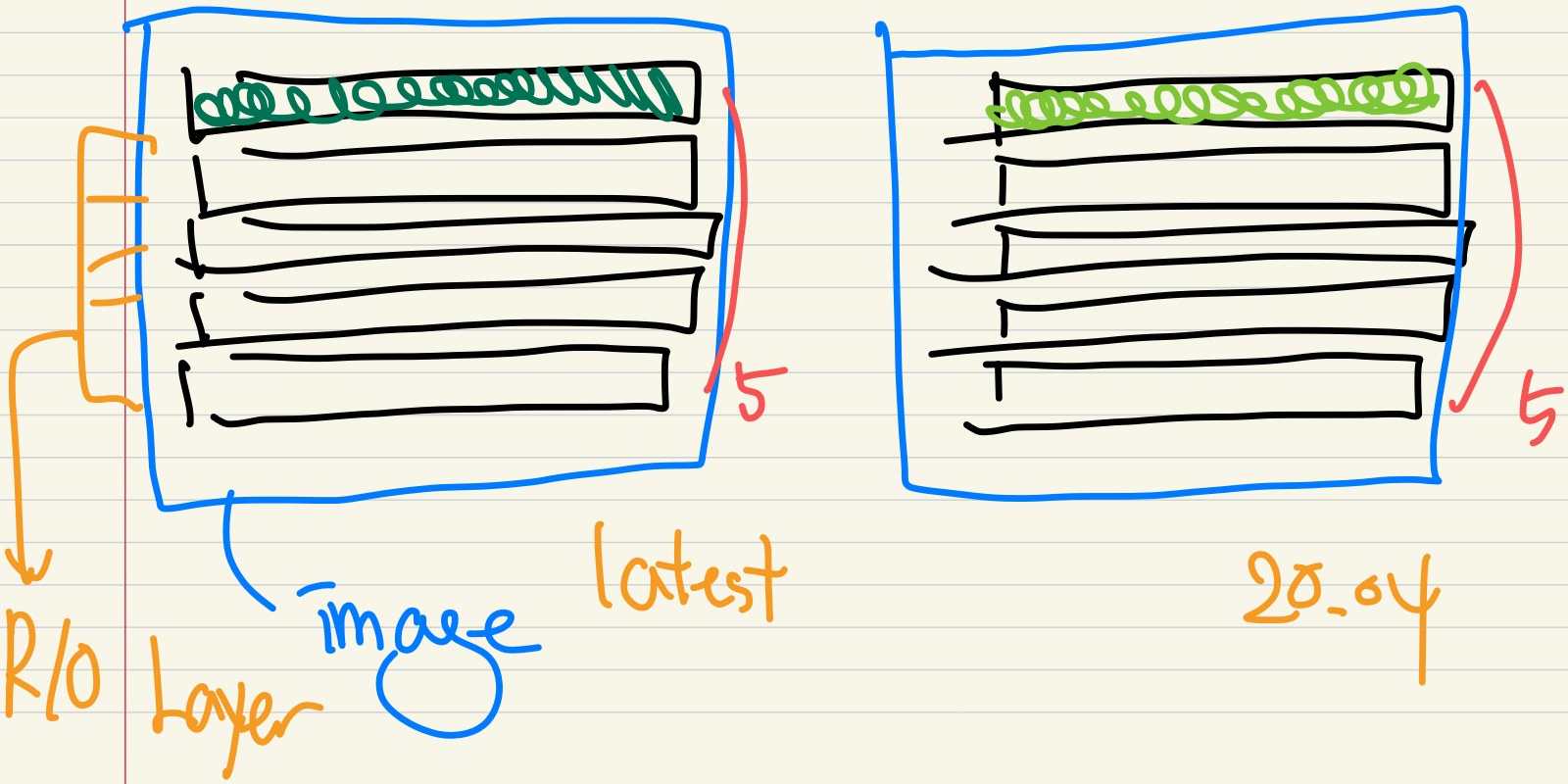
:16.04

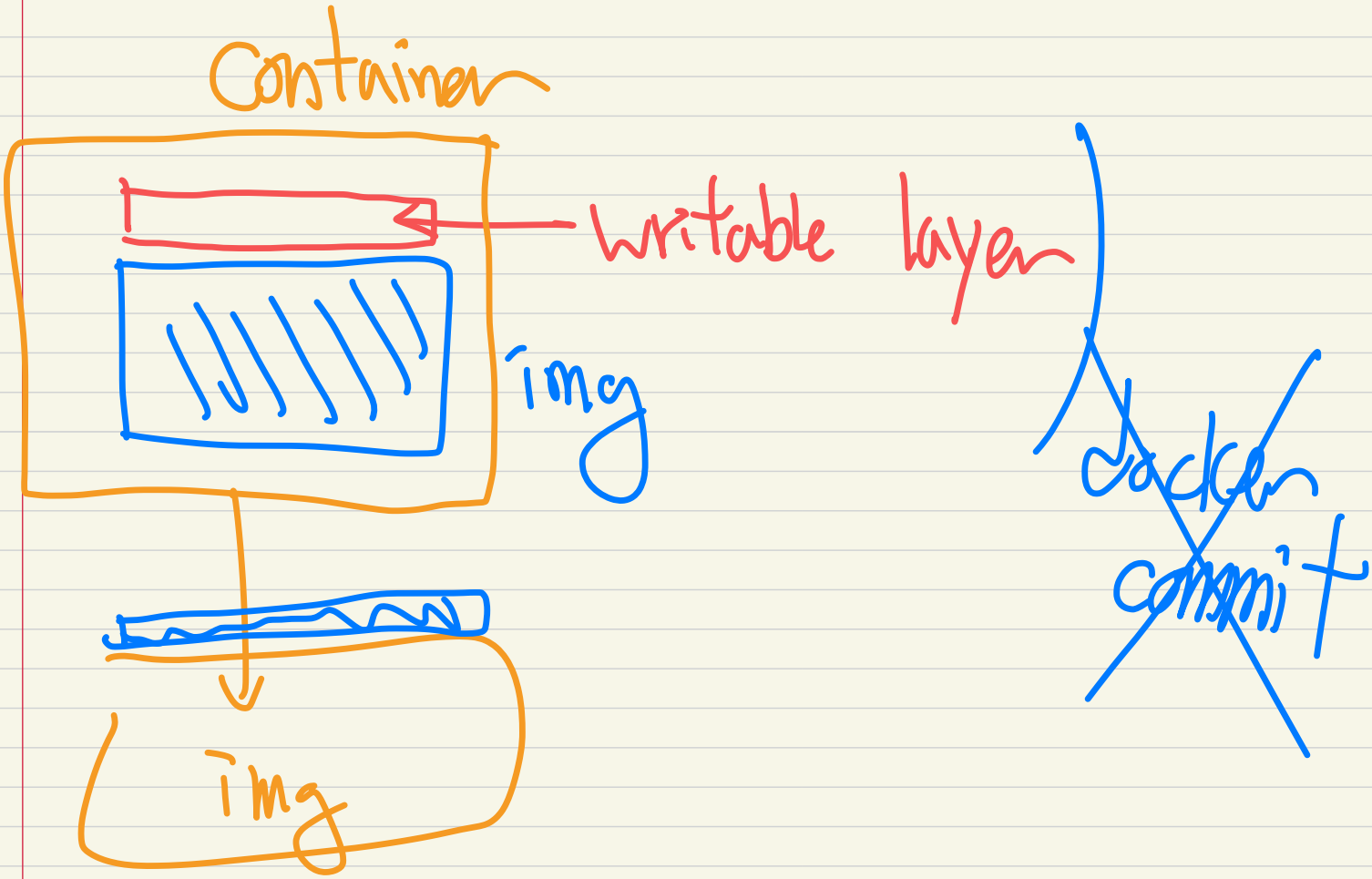
~~2GB~~

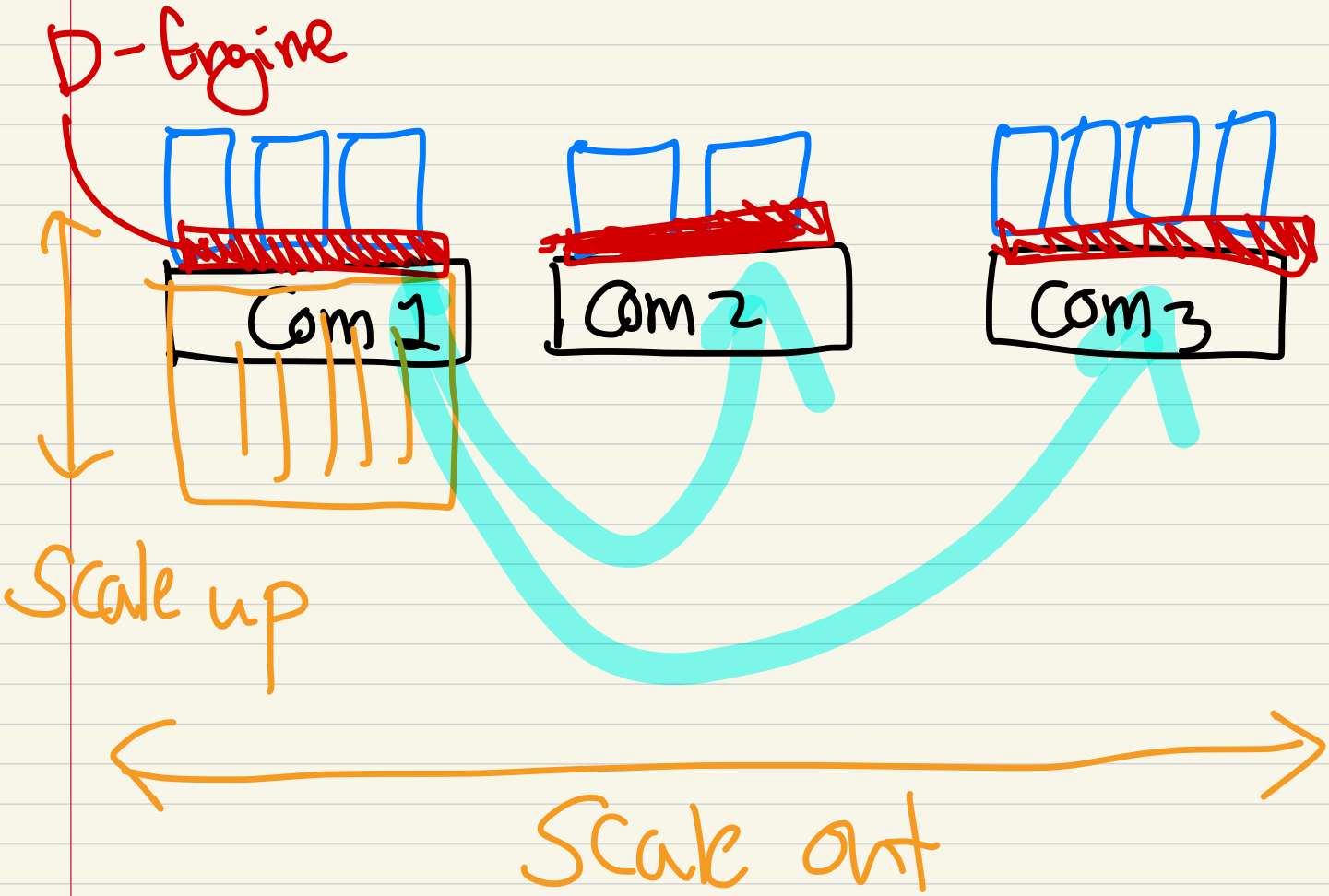
1.1GB

Img Layering

⇒ ~~10~~ 6




Container
writable layer
img
docker commit
img


D-Engine
Com 1
Com 2
Com 3
Scale up
Scale out

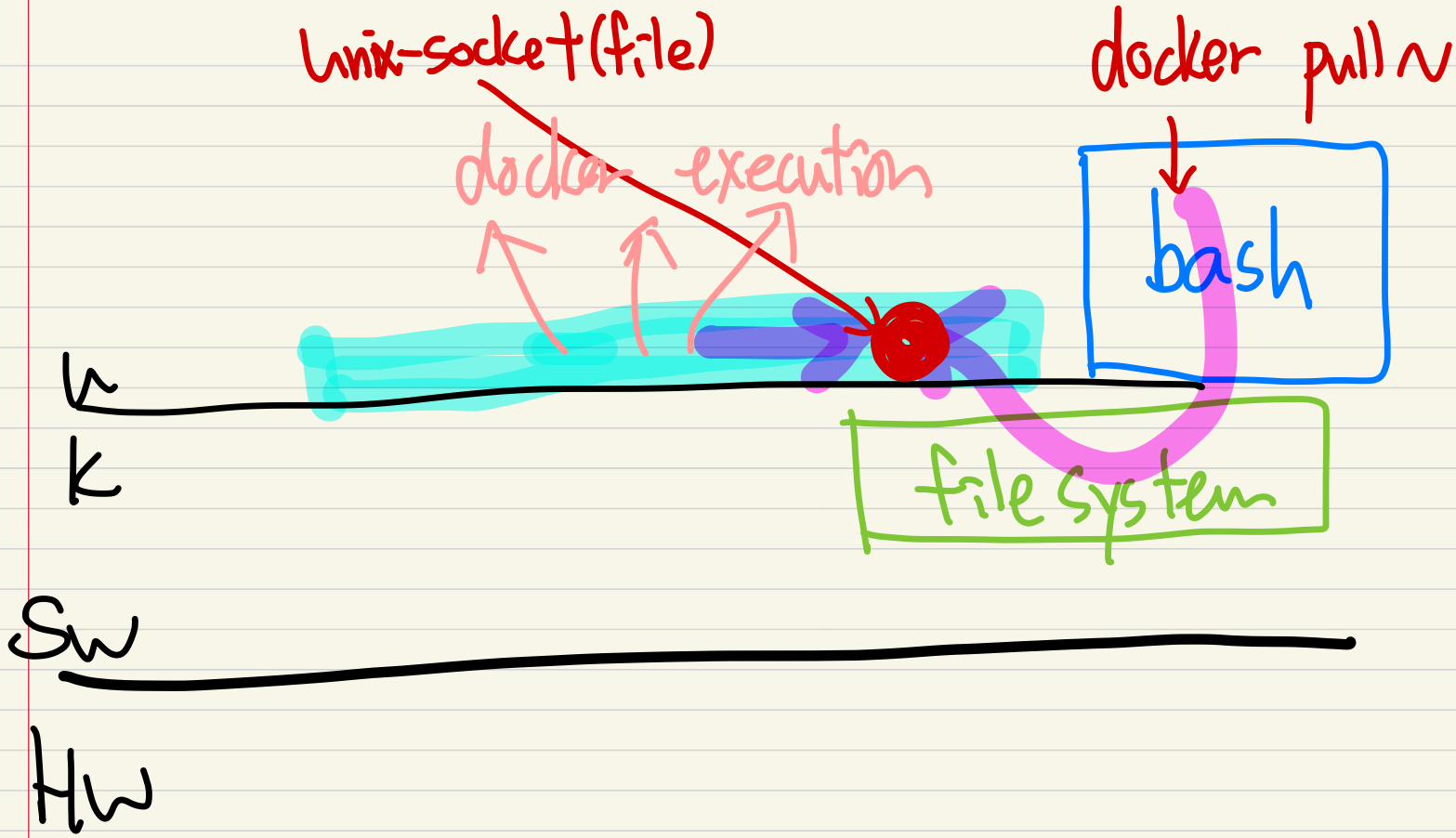
1. Docker Context (ssh & unix-socket)

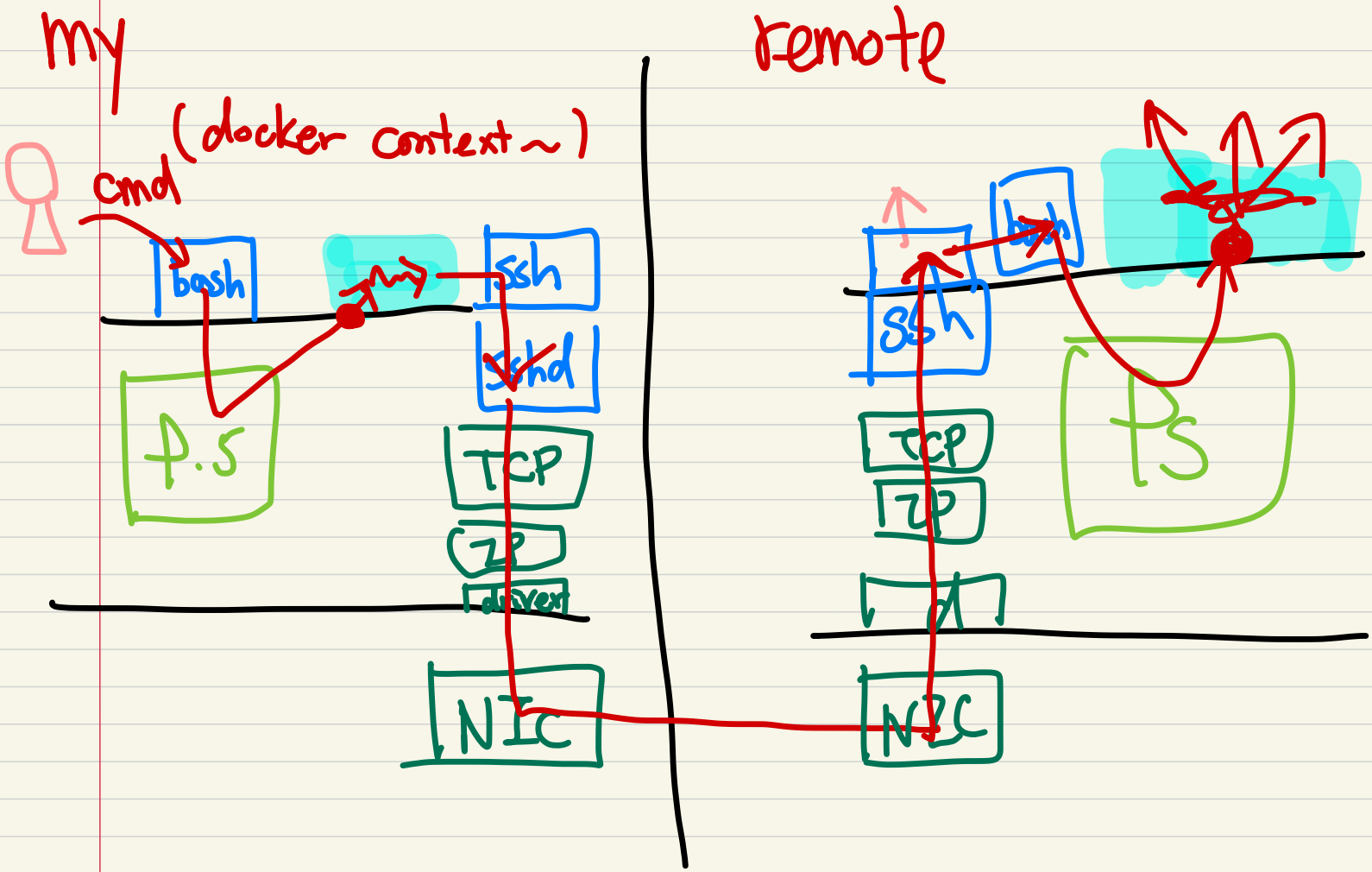
2-1. TCP-socket (2375/tcp)

~~2. Cont-Orch-tool~~

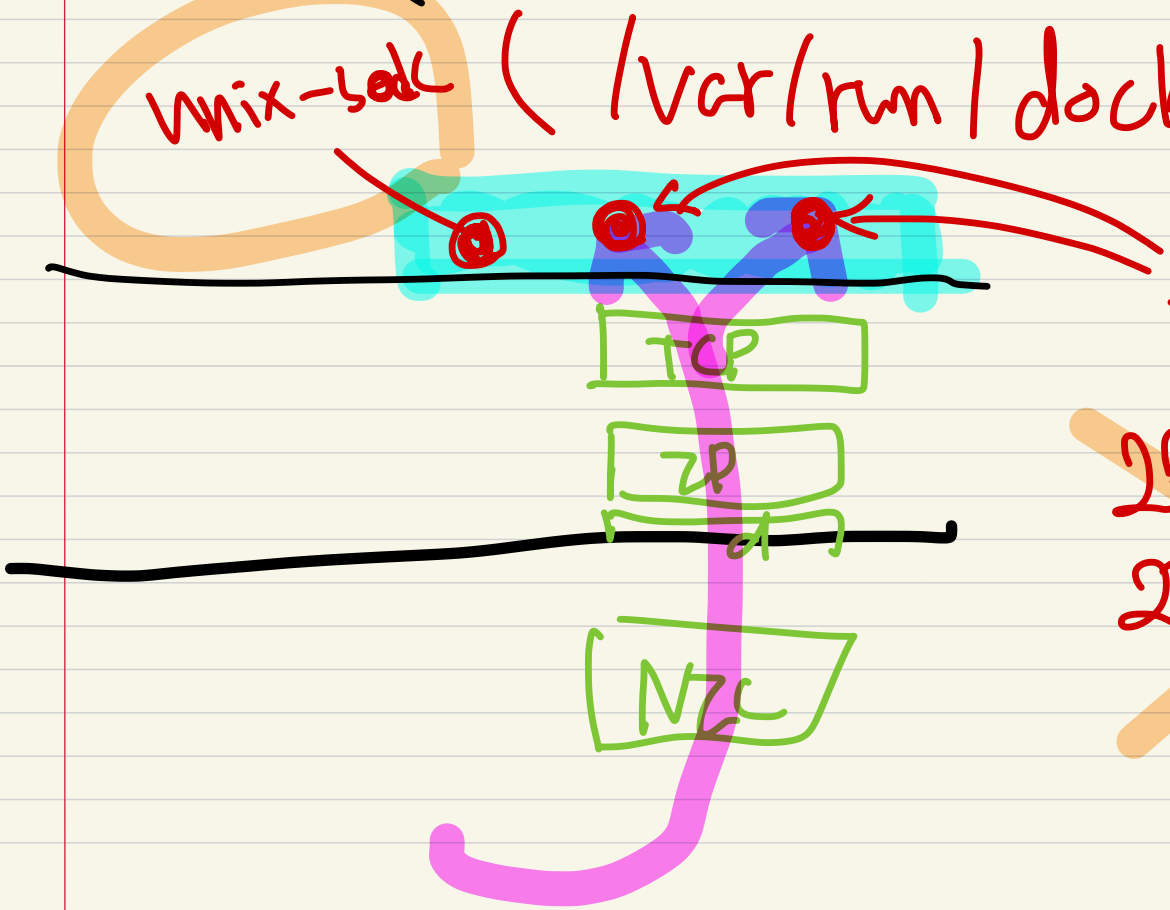
~~- docker-swarm~~

~~- K8s~~


unix-socket(file)
docker pull ~
docker execution
bash
K
file system
SW
HW


my
remote
(docker context ~)
cmd
bash
ssh
sshd
P.S
TCP
IP
ethernet
NIC
ssh
bash
TCP
IP
PS
NIC

2-1. TCP (TLS)



TCP-sock

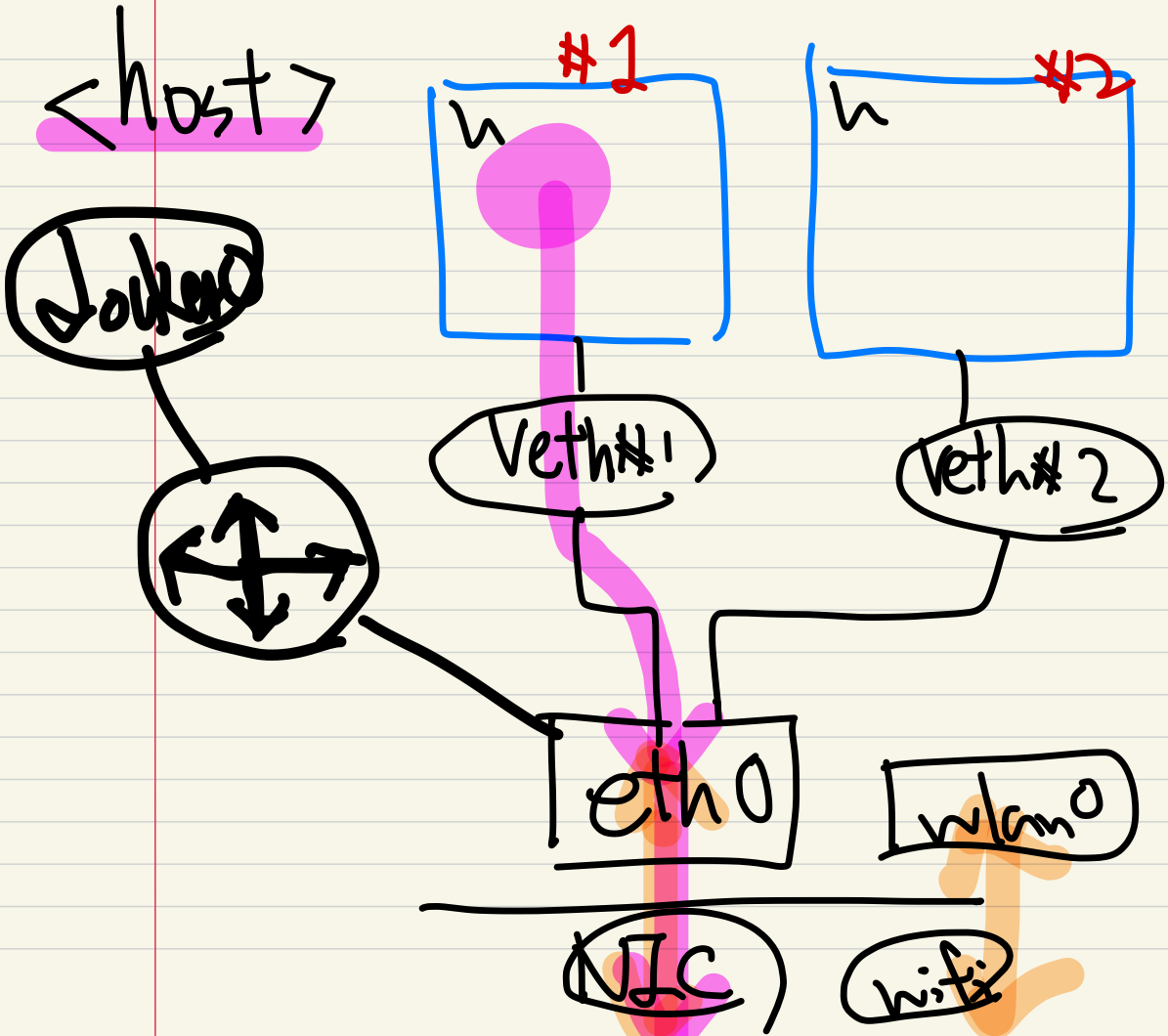
~~2375 - HTTP~~

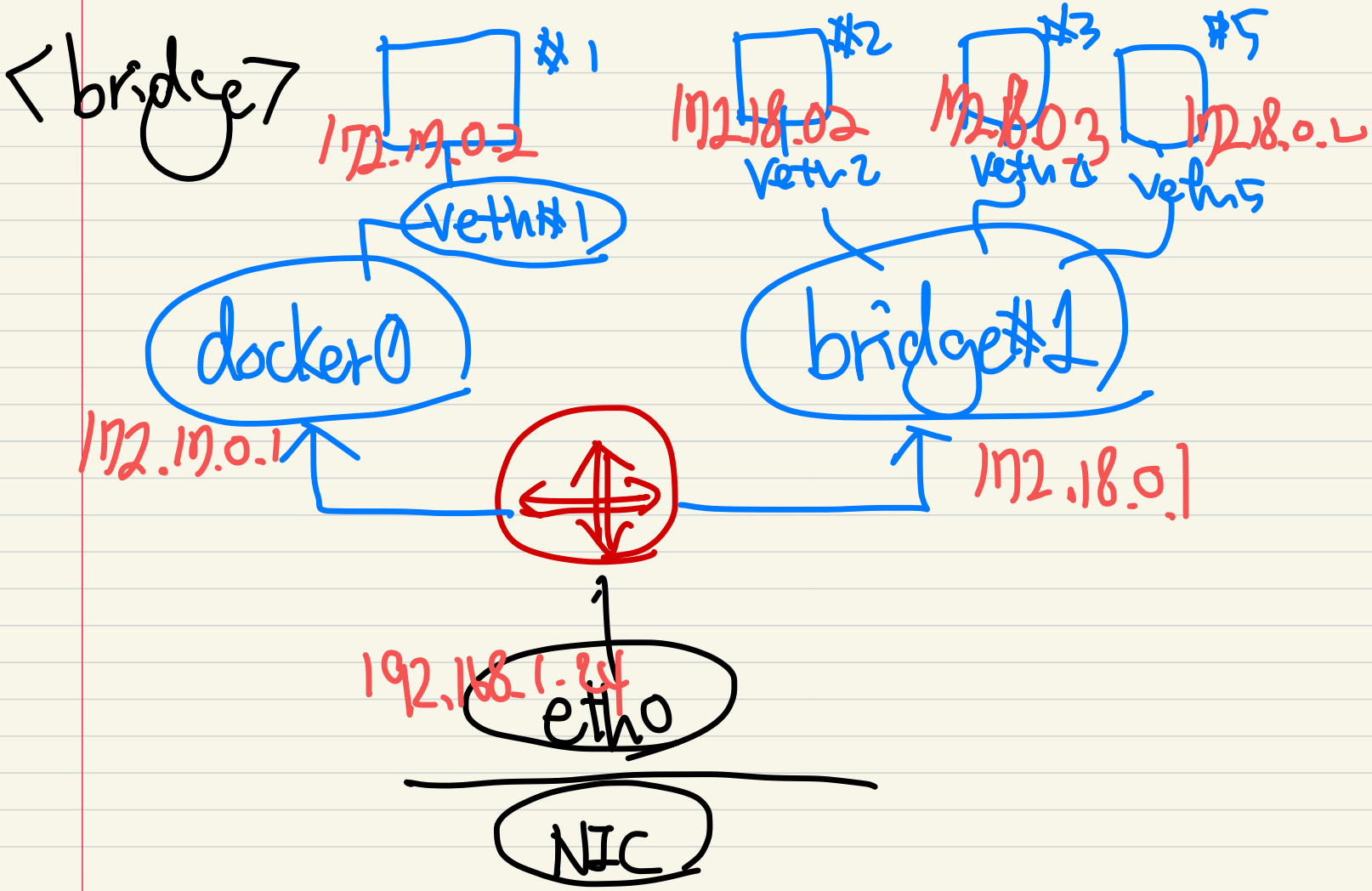
~~2376 - HTTPs (TLS)~~

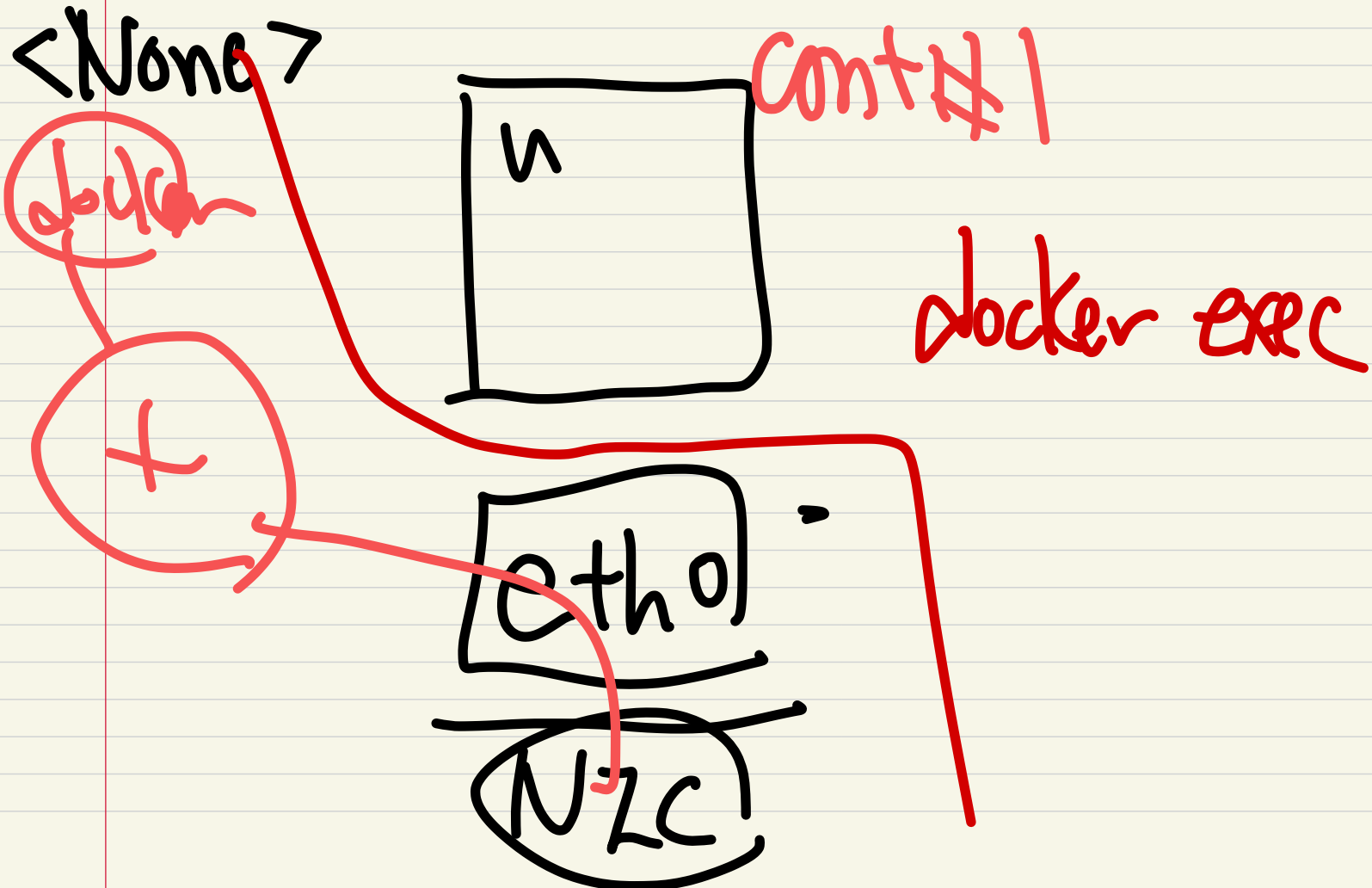
Docker-Network

- host
- ☆ bridge (docker0)
- ~~macvlan~~
- ~~ipvlan~~

- ~~none~~ ?


<host>
#1
#2
h
h
Jokeno
veth#1
veth#2
eth0
wlan0
NIC
wifi


<bridge>
#1
#2
#3
#5
172.17.0.2
172.18.0.2
172.18.0.3
172.18.0.5
veth#1
veth2
veth3
veth5
docker0
bridge#1
172.17.0.1
172.18.0.1
192.168.1.84
eth0
NIC


<None>
docker
+
n
cont#1
docker exec
eth0
NZC